DSP_POWER

File: DSP_POWER.kicad_sch

DSP-I

File: DSP-I.kicad_sch

DSP-II

File: DSP-II.kicad_sch

Serial_SATA

File: Serial_SATA.kicad_sch

SD_Card_LCD

File: SD_Card_LCD.kicad_sch

Video_Decoder (page 12)

File: Video_Decoder.kicad_sch

Power_Supply-I

File: Power_Supply-I.kicad_sch

Power_Supply_II

File: Power_Supply_II.kicad_sch

LCD_II

File: LCD_II.kicad_sch

DDR_RAM

File: DDR_RAM.kicad_sch

NAND_Flash

File: NAND_Flash.kicad_sch

Sound_Coded

File: Sound_Coded.kicad_sch

LAN_8710A

File: LAN_8710A.kicad_sch

Connectors

File: Connectors.kicad_sch

Mislleneous

File: Mislleneous.kicad_sch

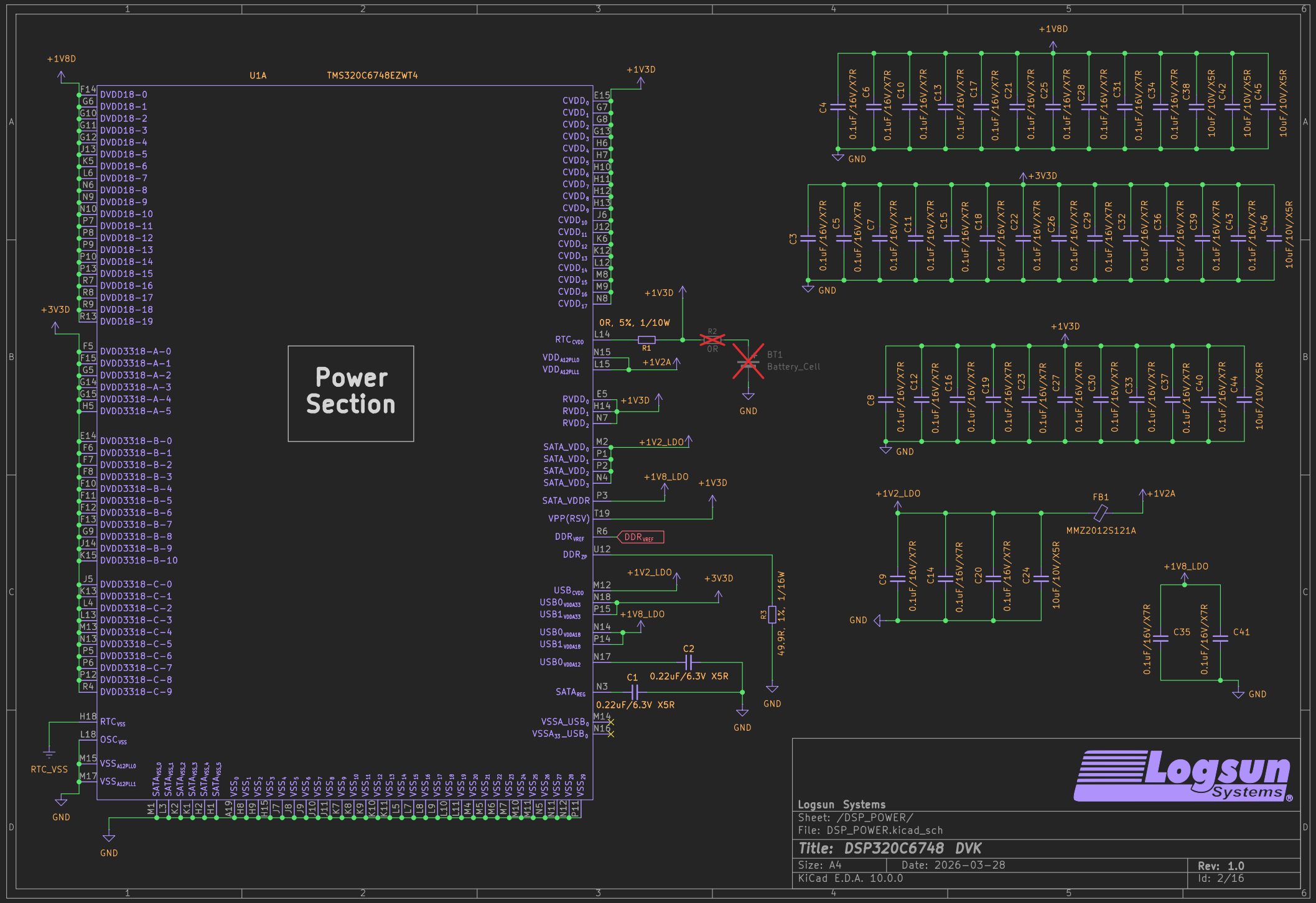
Logsun Systems

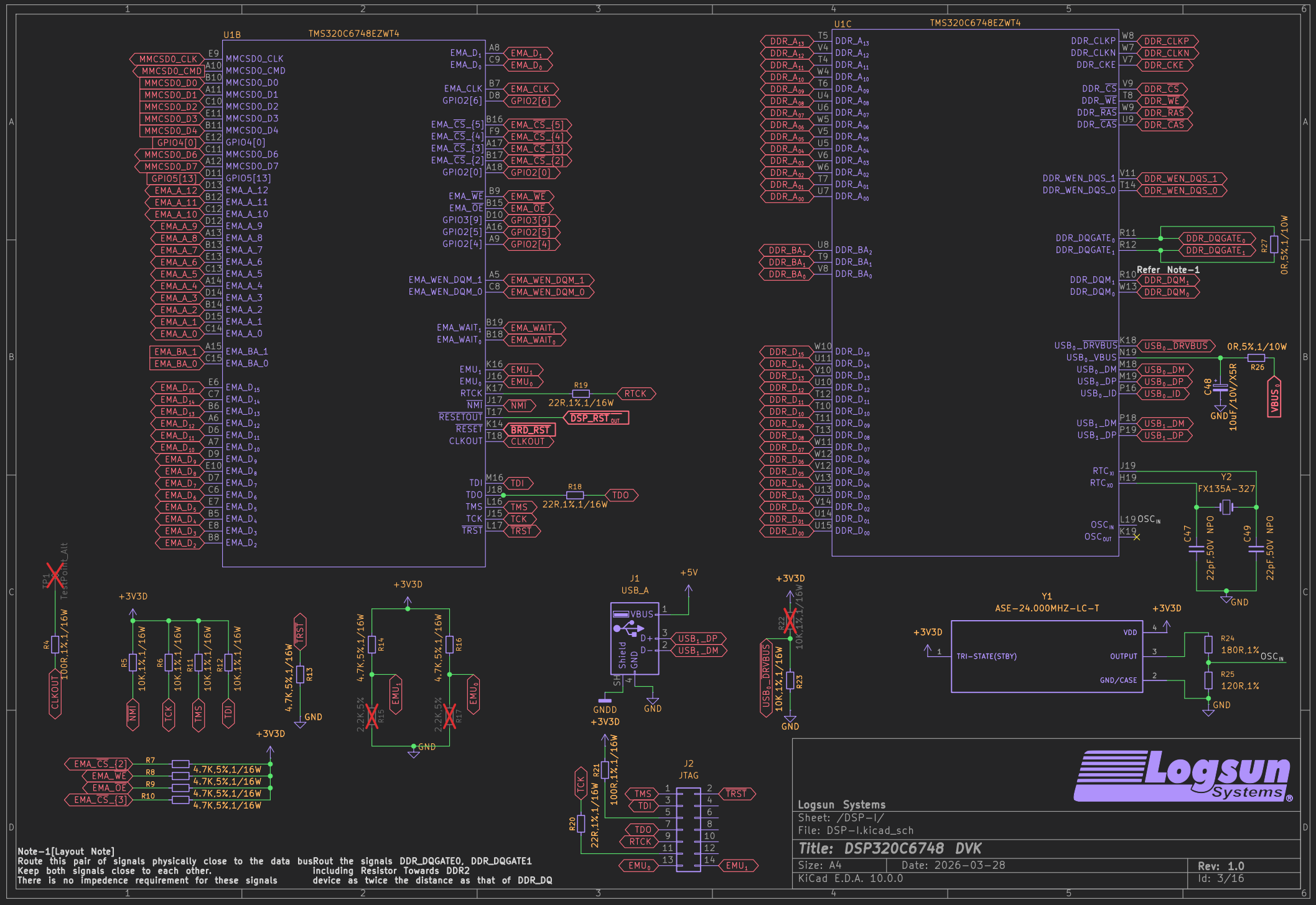
Sheet: /

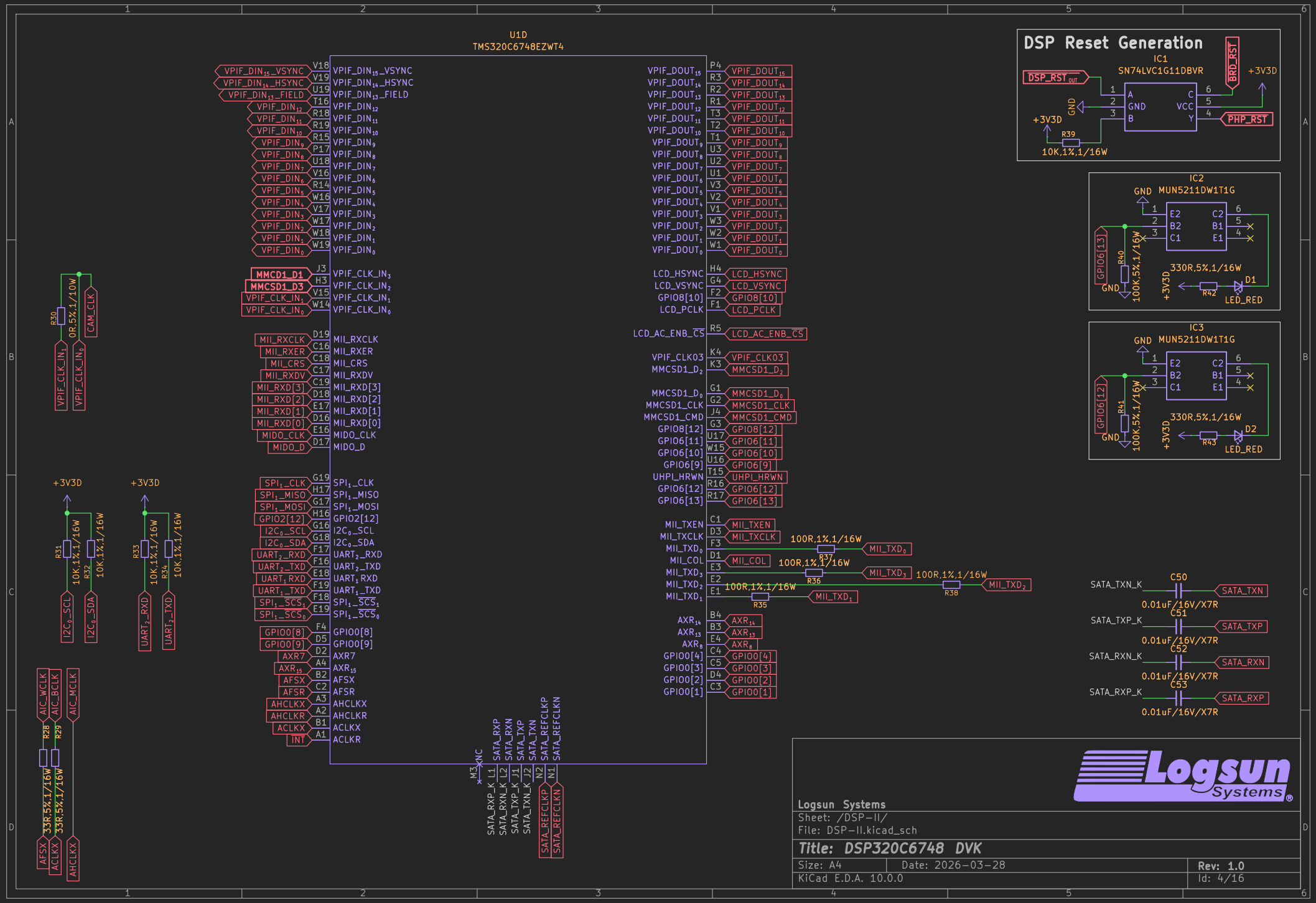
File: DSP320C6748_DVK.kicad_sch

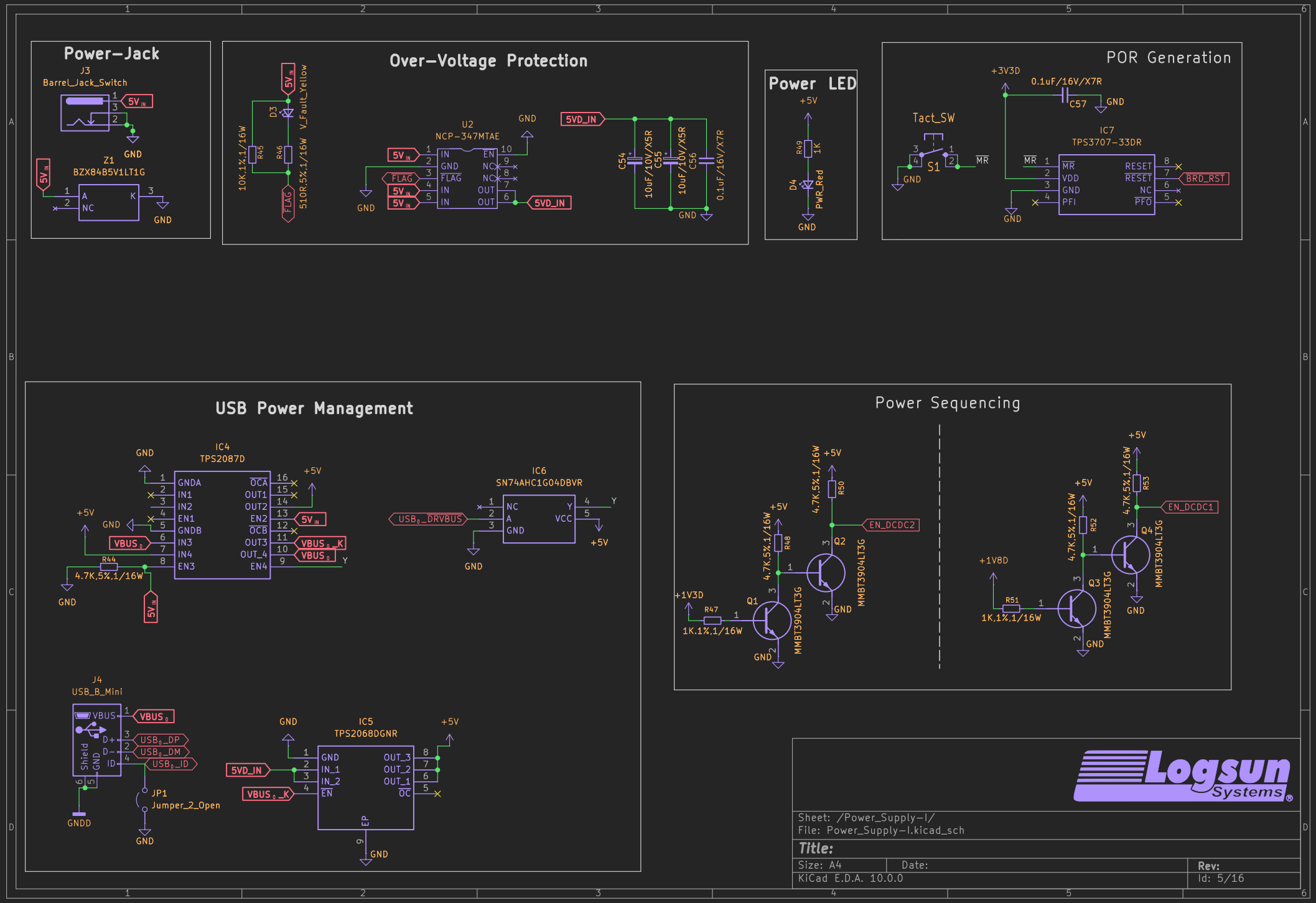
Title: DSP320C6748 DVK

Size: A4	Date: 2026-03-28	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 1/16


U1A
TMS320C6748EZWT4
Power Section
+1V8D
+3V3D
+1V3D
+1V2A
+1V2_LDO
+1V8_LDO
GND
RTC_VSS
OR, 5%, 1/10W
R1
R2
OR
BT1
Battery_Cell
DDR_VREF
0.22uF/6.3V X5R
C1
C2
R3
49.9R, 1%, 1/16W
FB1
MMZ2012S121A
Logsun Systems
Sheet: /DSP_POWER/
File: DSP_POWER.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 2/16


U1B
TMS320C6748EZWT4
U1C
TMS320C6748EZWT4
MMCSD0_CLK
MMCSD0_CMD
MMCSD0_D0
MMCSD0_D1
MMCSD0_D2
MMCSD0_D3
MMCSD0_D4
GPIO4[0]
MMCSD0_D6
MMCSD0_D7
GPIO5[13]
EMA_A_12
EMA_A_11
EMA_A_10
EMA_A_9
EMA_A_8
EMA_A_7
EMA_A_6
EMA_A_5
EMA_A_4
EMA_A_3
EMA_A_2
EMA_A_1
EMA_A_0
EMA_BA_1
EMA_BA_0
EMA_D15
EMA_D14
EMA_D13
EMA_D12
EMA_D11
EMA_D10
EMA_D9
EMA_D8
EMA_D7
EMA_D6
EMA_D5
EMA_D4
EMA_D3
EMA_D2
EMA_D1
EMA_D0
EMA_CLK
GPIO2[6]
EMA_CS_{5}
EMA_CS_{4}
EMA_CS_{3}
EMA_CS_{2}
GPIO2[0]
EMA_WE
EMA_OE
GPIO3[9]
GPIO2[5]
GPIO2[4]
EMA_WEN_DQM_1
EMA_WEN_DQM_0
EMA_WAIT1
EMA_WAIT0
EMU1
EMU0
RTCK
NMI
RESETOUT
RESET
CLKOUT
TDI
TDO
TMS
TCK
TRST
R19
22R,1%,1/16W
DSP_RST
BRD_RST
R18
22R,1%,1/16W
DDR_A13
DDR_A12
DDR_A11
DDR_A10
DDR_A09
DDR_A08
DDR_A07
DDR_A06
DDR_A05
DDR_A04
DDR_A03
DDR_A02
DDR_A01
DDR_A00
DDR_BA2
DDR_BA1
DDR_BA0
DDR_D15
DDR_D14
DDR_D13
DDR_D12
DDR_D11
DDR_D10
DDR_D09
DDR_D08
DDR_D07
DDR_D06
DDR_D05
DDR_D04
DDR_D03
DDR_D02
DDR_D01
DDR_D00
DDR_CLKP
DDR_CLKN
DDR_CKE
DDR_CS
DDR_WE
DDR_RAS
DDR_CAS
DDR_WEN_DQS_1
DDR_WEN_DQS_0
DDR_DQGATE0
DDR_DQGATE1
R27
0R,5%,1/10W
Refer Note-1
DDR_DQM1
DDR_DQM0
USB0_DRVBUS
USB0_VBUS
USB0_DM
USB0_DP
USB0_ID
USB1_DM
USB1_DP
R26
0R,5%,1/10W
C48
10uF/10V/X5R
GND
VBUS0
RTCXI
RTCXO
Y2
FX135A-327
OSCIN
OSCOUT
C47
22pF,50V NPO
C49
22pF,50V NPO
GND
TP1
TestPoint_Alt
R4
100R,1%,1/16W
CLKOUT
+3V3D
R5
10K,1%,1/16W
R6
10K,1%,1/16W
R11
10K,1%,1/16W
R12
10K,1%,1/16W
NMI
TCK
TMS
TDI
TRST
R13
4.7K,5%,1/16W
GND
R14
4.7K,5%,1/16W
R15
2.2K,5%
EMU1
R16
4.7K,5%,1/16W
R17
2.2K,5%
EMU0
J1
USB_A
+5V
VBUS
D+
D-
Shield
GND
SH
USB1_DP
USB1_DM
GNDD
R22
10K,1%,1/16W
USB0_DRVBUS
10K,1%,1/16W
R23
Y1
ASE-24.000MHZ-LC-T
TRI-STATE(STBY)
VDD
OUTPUT
GND/CASE
R24
180R,1%
R25
120R,1%
EMA_CS_{2}
EMA_WE
EMA_OE
EMA_CS_{3}
R7
R8
R9
R10
4.7K,5%,1/16W
R21
100R,1%,1/16W
R20
22R,1%,1/16W
J2
JTAG
TMS
TDI
TDO
RTCK
EMU0
EMU1
TRST
Logsun Systems
Sheet: /DSP-I/
File: DSP-I.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 3/16
Note-1[Layout Note]
Route this pair of signals physically close to the data bus
Keep both signals close to each other.
There is no impedence requirement for these signals
Rout the signals DDR_DQGATE0, DDR_DQGATE1
including Resistor Towards DDR2
device as twice the distance as that of DDR_DQ


U1D
TMS320C6748EZWT4
DSP Reset Generation
IC1
SN74LVC1G11DBVR
DSP_RST_OUT
BRD_RST
PHP_RST
+3V3D
GND
R39
10K,1%,1/16W
IC2
MUN5211DW1T1G
GPIO6[13]
R40
100K,5%,1/16W
330R,5%,1/16W
R42
D1
LED_RED
IC3
MUN5211DW1T1G
GPIO6[12]
R41
R43
D2
LED_RED
R30
0R,5%,1/10W
CAM_CLK
VPIF_CLK_IN1
VPIF_CLK_IN0
R31
R32
10K,1%,1/16W
I2C0_SCL
I2C0_SDA
R33
R34
UART2_RXD
UART2_TXD
AIC_WCLK
AIC_BCLK
AIC_MCLK
R28
R29
33R,5%,1/16W
AFSX
ACLKX
AHCLKX
MMCD1_D1
MMCSD1_D3
MII_TXEN
MII_TXCLK
MII_TXD0
MII_COL
MII_TXD3
MII_TXD2
MII_TXD1
R37
R36
R35
R38
100R,1%,1/16W
SATA_TXN_K
SATA_TXP_K
SATA_RXN_K
SATA_RXP_K
C50
C51
C52
C53
0.01uF/16V/X7R
SATA_TXN
SATA_TXP
SATA_RXN
SATA_RXP
SATA_REFCLKP
SATA_REFCLKN
NC
Logsun Systems
Sheet: /DSP-II/
File: DSP-II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 4/16


Power-Jack
J3
Barrel_Jack_Switch
5V_IN
GND
Z1
BZX84B5V1LT1G
A
K
NC
Over-Voltage Protection
D3
V_Fault_Yellow
10K,1%,1/16W
R45
R46
510R,5%,1/16W
FLAG
U2
NCP-347MTAE
IN
GND
FLAG
EN
NC
OUT
5VD_IN
C54
C55
C56
10uF/10V/X5R
0.1uF/16V/X7R
Power LED
+5V
R49
1K
D4
PWR_Red
POR Generation
+3V3D
0.1uF/16V/X7R
C57
Tact_SW
S1
MR
IC7
TPS3707-33DR
VDD
PFI
RESET
PFO
BRD_RST
USB Power Management
IC4
TPS2087D
GNDA
IN1
IN2
EN1
GNDB
IN3
IN4
EN3
OCA
OUT1
OUT2
EN2
OCB
OUT3
OUT_4
EN4
VBUS_0
VBUS_0_K
R44
4.7K,5%,1/16W
IC6
SN74AHC1G04DBVR
NC
A
GND
Y
VCC
USB_0_DRVBUS
J4
USB_B_Mini
VBUS
D+
D-
ID
Shield
GND
USB_0_DP
USB_0_DM
USB_0_ID
JP1
Jumper_2_Open
GNDD
IC5
TPS2068DGNR
GND
IN_1
IN_2
EN
OUT_3
OUT_2
OUT_1
OC
EP
Power Sequencing
R50
EN_DCDC2
Q2
MMBT3904LT3G
R48
Q1
R47
1K,1%,1/16W
+1V3D
R53
EN_DCDC1
Q4
R52
Q3
R51
1K,1%,1/16W
+1V8D
Logsun Systems
Sheet: /Power_Supply-1/
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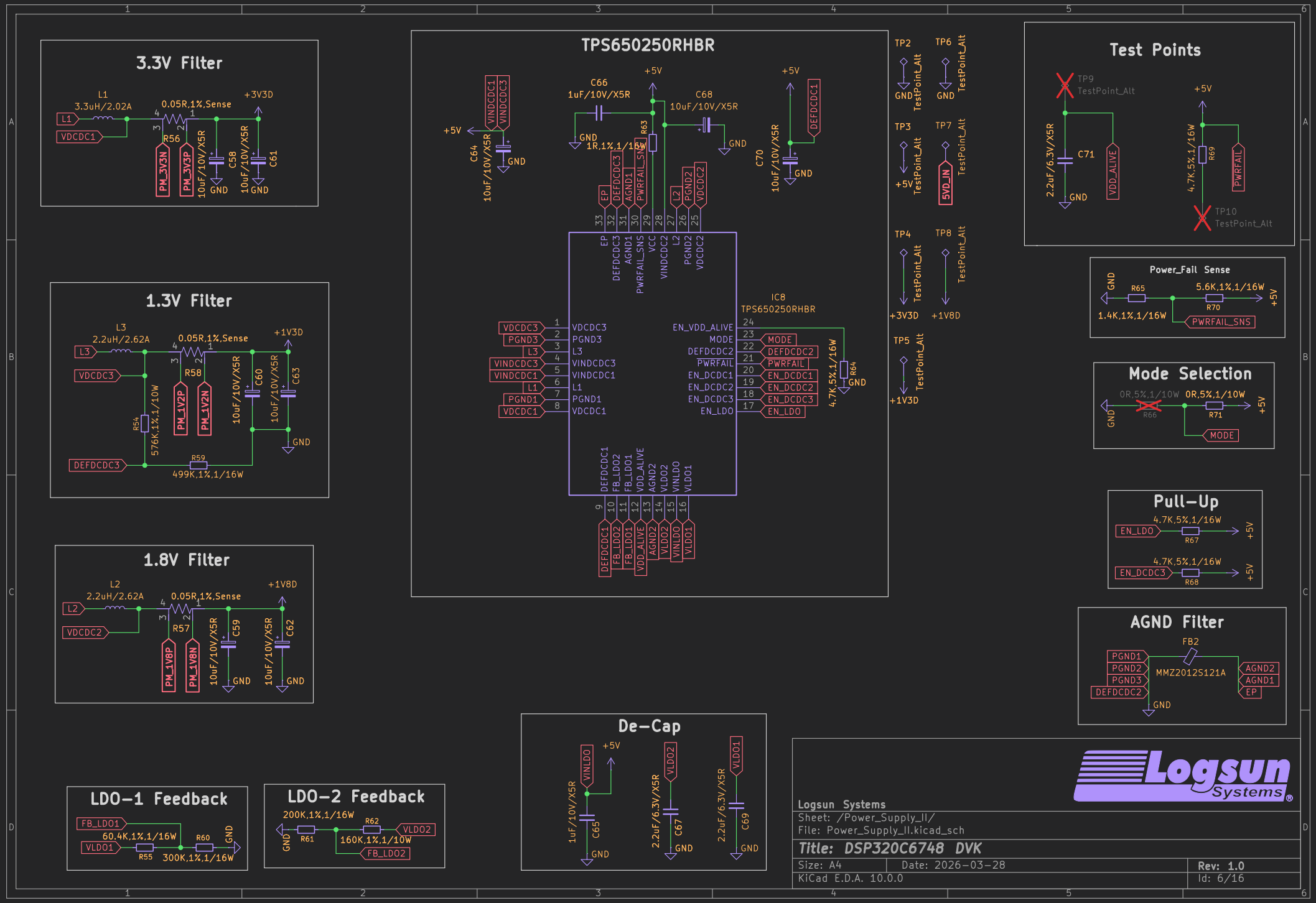
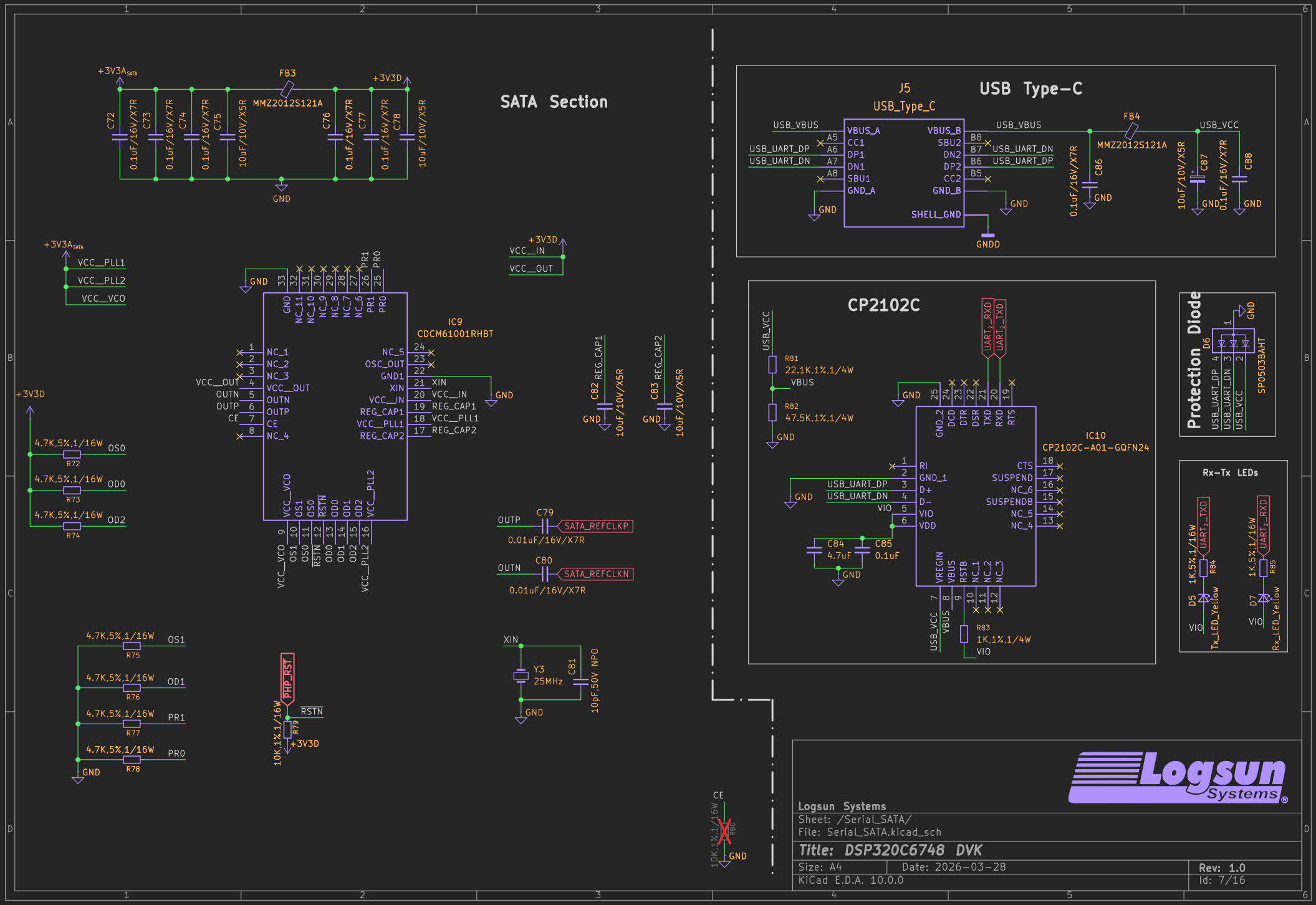
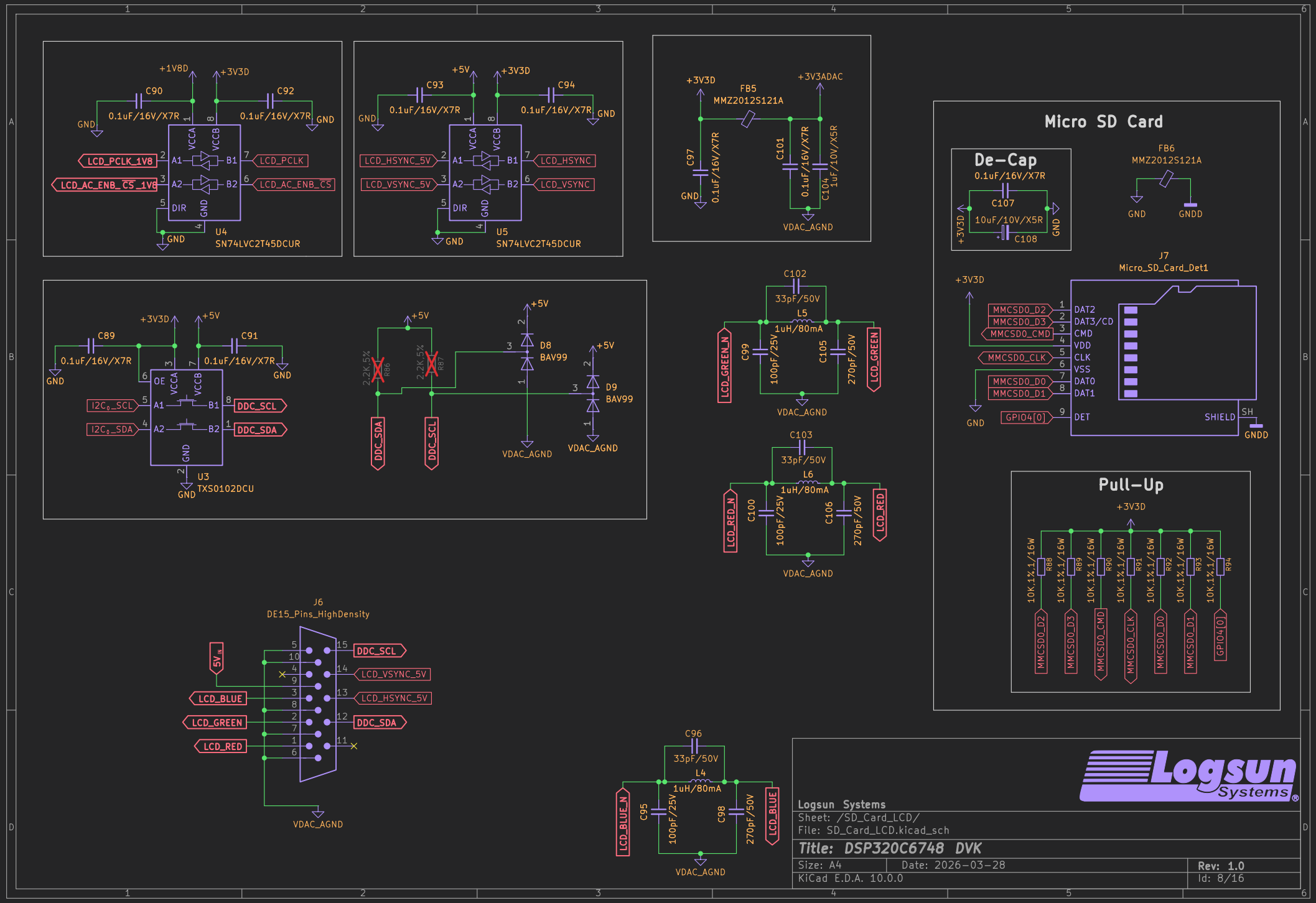
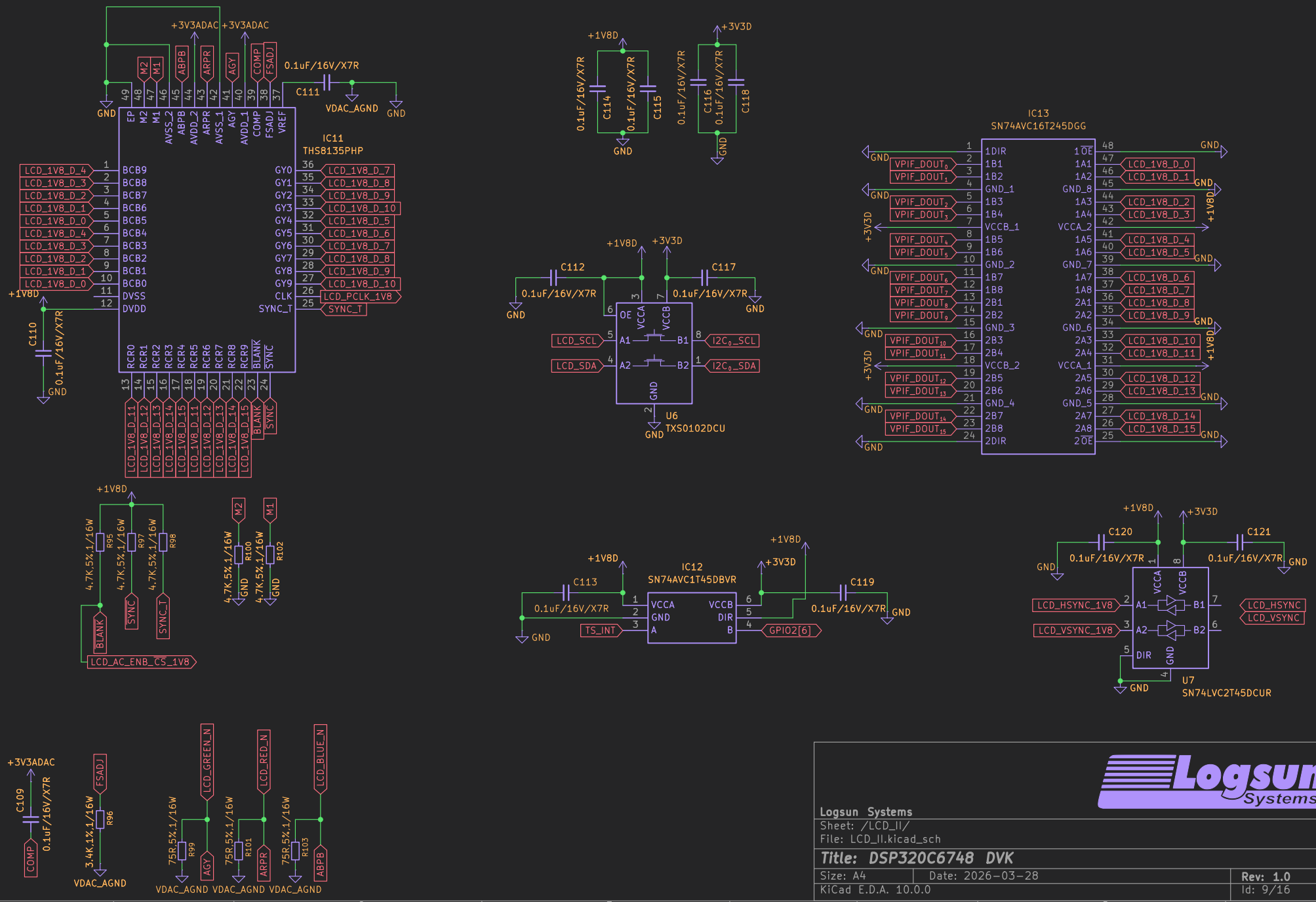
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 5/16


3.3V Filter
L1
3.3uH/2.02A
0.05R,1%,Sense
+3V3D
L1
VDCDC1
R56
PM_3V3N
PM_3V3P
10uF/10V/X5R
C58
C61
GND
1.3V Filter
L3
2.2uH/2.62A
0.05R,1%,Sense
+1V3D
L3
VDCDC3
R58
PM_1V2P
PM_1V2N
C60
C63
R54
576K,1%,1/10W
DEFDCDC3
R59
499K,1%,1/16W
1.8V Filter
L2
2.2uH/2.62A
0.05R,1%,Sense
+1V8D
L2
VDCDC2
R57
PM_1V8P
PM_1V8N
C59
C62
TPS650250RHBR
VINDCDC1
VINDCDC3
C64
C66
1uF/10V/X5R
C68
10uF/10V/X5R
R63
1R,1%,1/16W
C70
DEFDCDC1
EP
DEFDCDC3
AGND1
PWRFAIL_SNS
VCC
VINDCDC2
L2
PGND2
VDCDC2
IC8
TPS650250RHBR
VDCDC3
PGND3
L3
VINDCDC3
VINDCDC1
L1
PGND1
VDCDC1
EN_VDD_ALIVE
MODE
DEFDCDC2
PWRFAIL
EN_DCDC1
EN_DCDC2
EN_DCDC3
EN_LDO
R64
4.7K,5%,1/16W
DEFDCDC1
FB_LDO2
FB_LDO1
VDD_ALIVE
AGND2
VLDO2
VINLDO
VLDO1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TestPoint_Alt
5VD_IN
+3V3D
+1V8D
+1V3D
Test Points
TP9
TP10
C71
2.2uF/6.3V/X5R
VDD_ALIVE
R69
4.7K,5%,1/16W
PWRFAIL
Power_Fail Sense
R65
1.4K,1%,1/16W
R70
5.6K,1%,1/16W
PWRFAIL_SNS
Mode Selection
0R,5%,1/10W
R66
R71
MODE
Pull-Up
EN_LDO
R67
4.7K,5%,1/16W
EN_DCDC3
R68
AGND Filter
FB2
PGND1
PGND2
PGND3
DEFDCDC2
MMZ2012S121A
AGND2
AGND1
EP
De-Cap
VINLDO
C65
1uF/10V/X5R
VLDO2
C67
2.2uF/6.3V/X5R
VLDO1
C69
2.2uF/6.3V/X5R
LDO-1 Feedback
FB_LDO1
VLDO1
60.4K,1%,1/16W
R55
R60
300K,1%,1/16W
LDO-2 Feedback
200K,1%,1/16W
R61
R62
160K,1%,1/10W
VLDO2
FB_LDO2
Logsun Systems
Sheet: /Power_Supply_II/
File: Power_Supply_II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 6/16


SATA Section
+3V3A_SATA
FB3
MMZ2012S121A
+3V3D
C72 0.1uF/16V/X7R
C73 0.1uF/16V/X7R
C74 0.1uF/16V/X7R
C75 0.1uF/16V/X7R
10uF/10V/X5R
C76 0.1uF/16V/X7R
C77 0.1uF/16V/X7R
C78 10uF/10V/X5R
GND
J5
USB_Type_C
USB Type-C
USB_VBUS
USB_UART_DP
USB_UART_DN
VBUS_A
CC1
DP1
DN1
SBU1
GND_A
VBUS_B
SBU2
DN2
DP2
CC2
GND_B
SHELL_GND
GNDD
FB4
MMZ2012S121A
USB_VCC
C86 0.1uF/16V/X7R
C87 10uF/10V/X5R
C88 0.1uF/16V/X7R
+3V3A_SATA
VCC__PLL1
VCC__PLL2
VCC__VCO
+3V3D
VCC__IN
VCC__OUT
IC9
CDCM61001RHBT
NC_1
NC_2
NC_3
VCC__OUT
OUTN
OUTP
CE
NC_4
NC_5
OSC_OUT
GND1
XIN
VCC__IN
REG_CAP1
VCC__PLL1
REG_CAP2
C82 REG_CAP1 10uF/10V/X5R
C83 REG_CAP2 10uF/10V/X5R
OUTP C79 SATA_REFCLKP 0.01uF/16V/X7R
OUTN C80 SATA_REFCLKN 0.01uF/16V/X7R
4.7K,5%,1/16W OS0 R72
4.7K,5%,1/16W OD0 R73
4.7K,5%,1/16W OD2 R74
4.7K,5%,1/16W OS1 R75
4.7K,5%,1/16W OD1 R76
4.7K,5%,1/16W PR1 R77
4.7K,5%,1/16W PR0 R78
PHP_RST
10K,1%,1/16W R79
RSTN
XIN
Y3 25MHz
C81 10pF,50V NPO
CP2102C
USB_VCC
R81 22.1K,1%,1/4W
R82 47.5K,1%,1/4W
VBUS
UART_2_RXD
UART_2_TXD
IC10
CP2102C-A01-GQFN24
RI
GND_1
D+
D-
VIO
VDD
VREGIN
RSTB
CTS
SUSPEND
NC_6
SUSPENDB
NC_5
NC_4
C84 4.7uF
C85 0.1uF
R83 1K,1%,1/4W
Protection Diode
D6 SP0503BAHT
USB_UART_DP
USB_UART_DN
USB_VCC
Rx-Tx LEDs
1K,5%,1/16W R84
D5 Tx_LED_Yellow
1K,5%,1/16W R85
D7 Rx_LED_Yellow
CE
10K,1%,1/16W R80
Logsun Systems
Sheet: /Serial_SATA/
File: Serial_SATA.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 7/16


+1V8D
+3V3D
C90
0.1uF/16V/X7R
C92
0.1uF/16V/X7R
GND
LCD_PCLK_1V8
LCD_AC_ENB_CS_1V8
VCCA
VCCB
A1
A2
B1
B2
DIR
LCD_PCLK
LCD_AC_ENB_CS
U4
SN74LVC2T45DCUR
+5V
C93
0.1uF/16V/X7R
C94
0.1uF/16V/X7R
LCD_HSYNC_5V
LCD_VSYNC_5V
LCD_HSYNC
LCD_VSYNC
U5
SN74LVC2T45DCUR
+3V3D
FB5
MMZ2012S121A
+3V3ADAC
C97
0.1uF/16V/X7R
C101
0.1uF/16V/X7R
C104
1uF/10V/X5R
VDAC_AGND
C89
0.1uF/16V/X7R
C91
0.1uF/16V/X7R
OE
I2C0_SCL
I2C0_SDA
DDC_SCL
DDC_SDA
U3
TXS0102DCU
2.2K 5%
R86
R87
D8
BAV99
D9
BAV99
C102
33pF/50V
L5
1uH/80mA
C99
100pF/25V
C105
270pF/50V
LCD_GREEN_N
LCD_GREEN
C103
33pF/50V
L6
1uH/80mA
C100
100pF/25V
C106
270pF/50V
LCD_RED_N
LCD_RED
J6
DE15_Pins_HighDensity
5V_IN
LCD_BLUE
LCD_GREEN
LCD_RED
DDC_SCL
LCD_VSYNC_5V
LCD_HSYNC_5V
DDC_SDA
C96
33pF/50V
L4
1uH/80mA
C95
100pF/25V
C98
270pF/50V
LCD_BLUE_N
LCD_BLUE
Micro SD Card
De-Cap
0.1uF/16V/X7R
C107
10uF/10V/X5R
C108
FB6
MMZ2012S121A
GNDD
J7
Micro_SD_Card_Det1
MMCSD0_D2
MMCSD0_D3
MMCSD0_CMD
MMCSD0_CLK
MMCSD0_D0
MMCSD0_D1
GPIO4[0]
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
SH
Pull-Up
10K,1%,1/16W
R88
R89
R90
R91
R92
R93
R94
Logsun Systems
Sheet: /SD_Card_LCD/
File: SD_Card_LCD.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 8/16

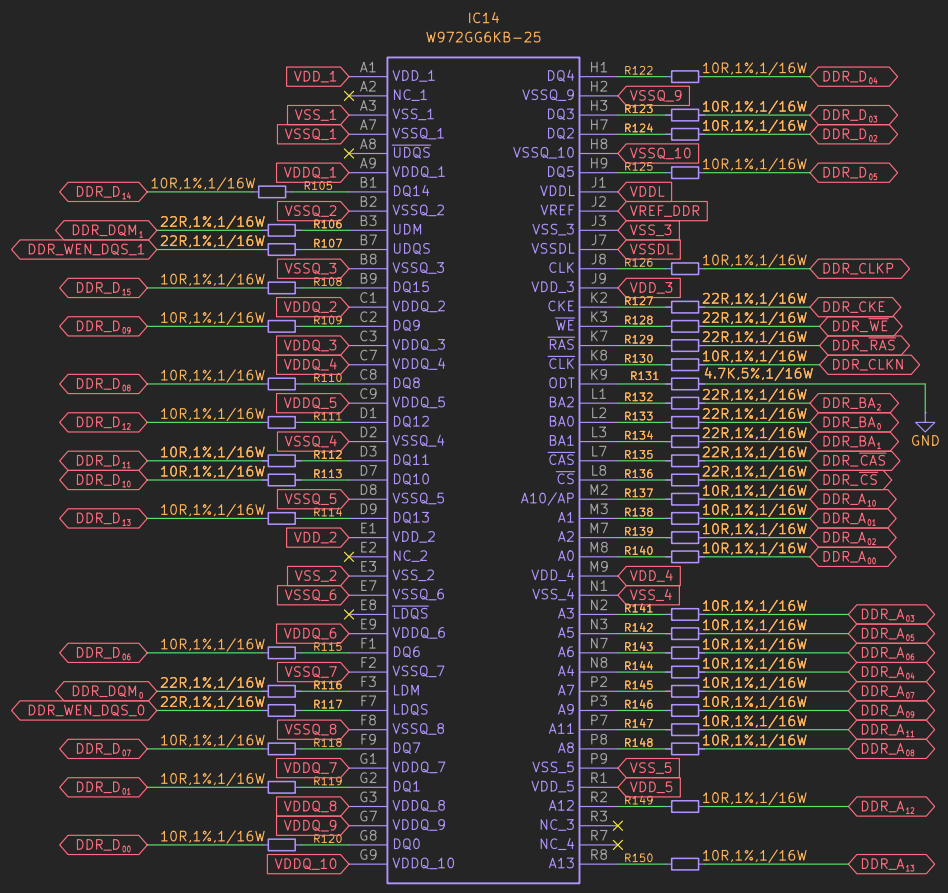


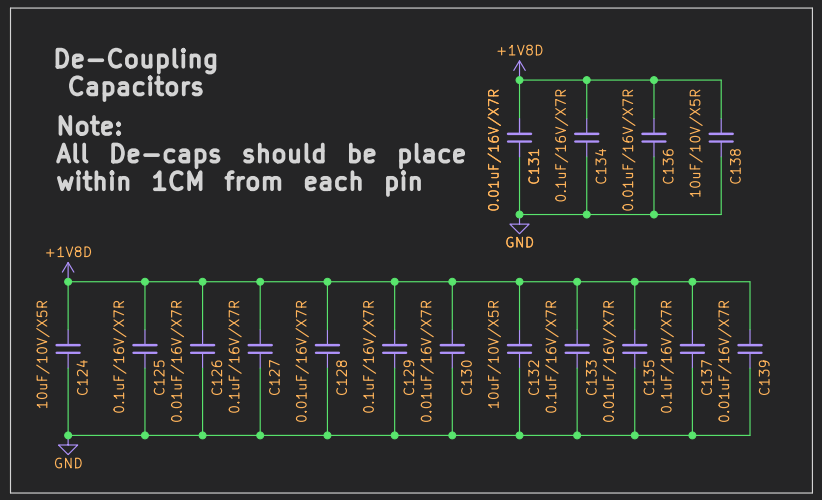
Logsun Systems

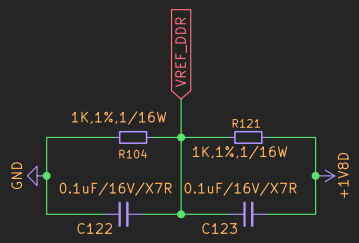
Sheet: /LCD_II/
File: LCD_II.kicad_sch

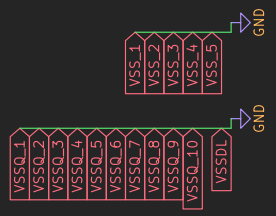
Title: DSP320C6748 DVK

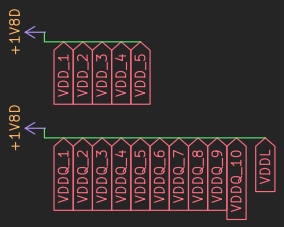
Size: A4	Date: 2026-03-28	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 9/16











Logsun Systems

Sheet: /DDR_RAM/
File: DDR_RAM.kicad_sch

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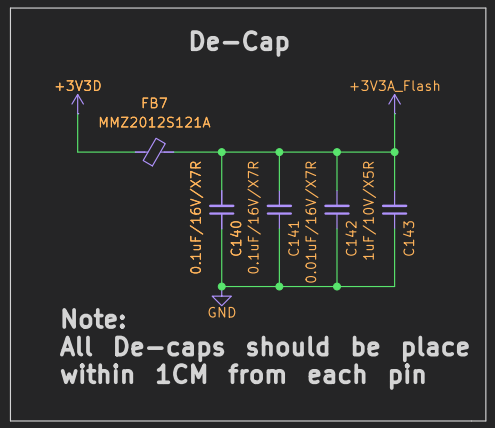
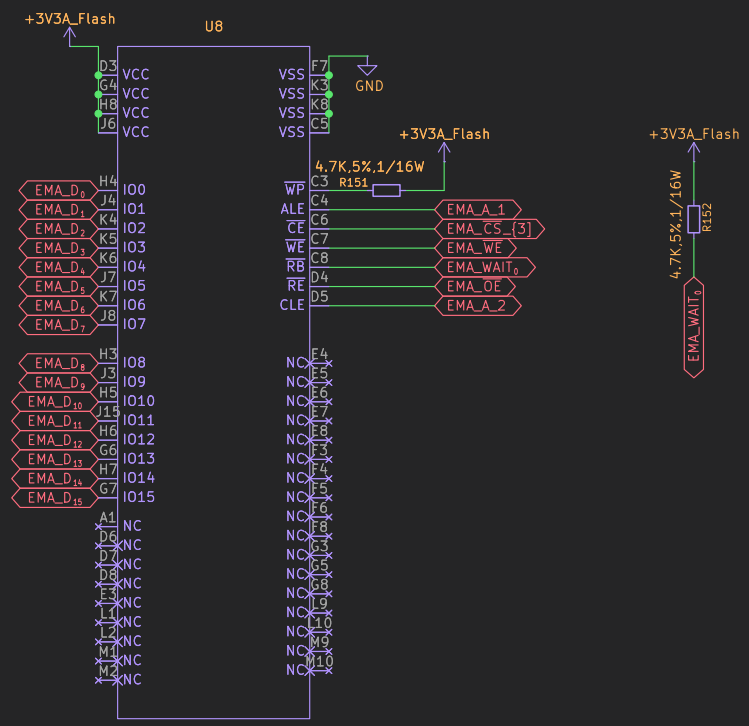
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 10/16





Logsun Systems

Sheet: /NAND_Flash/
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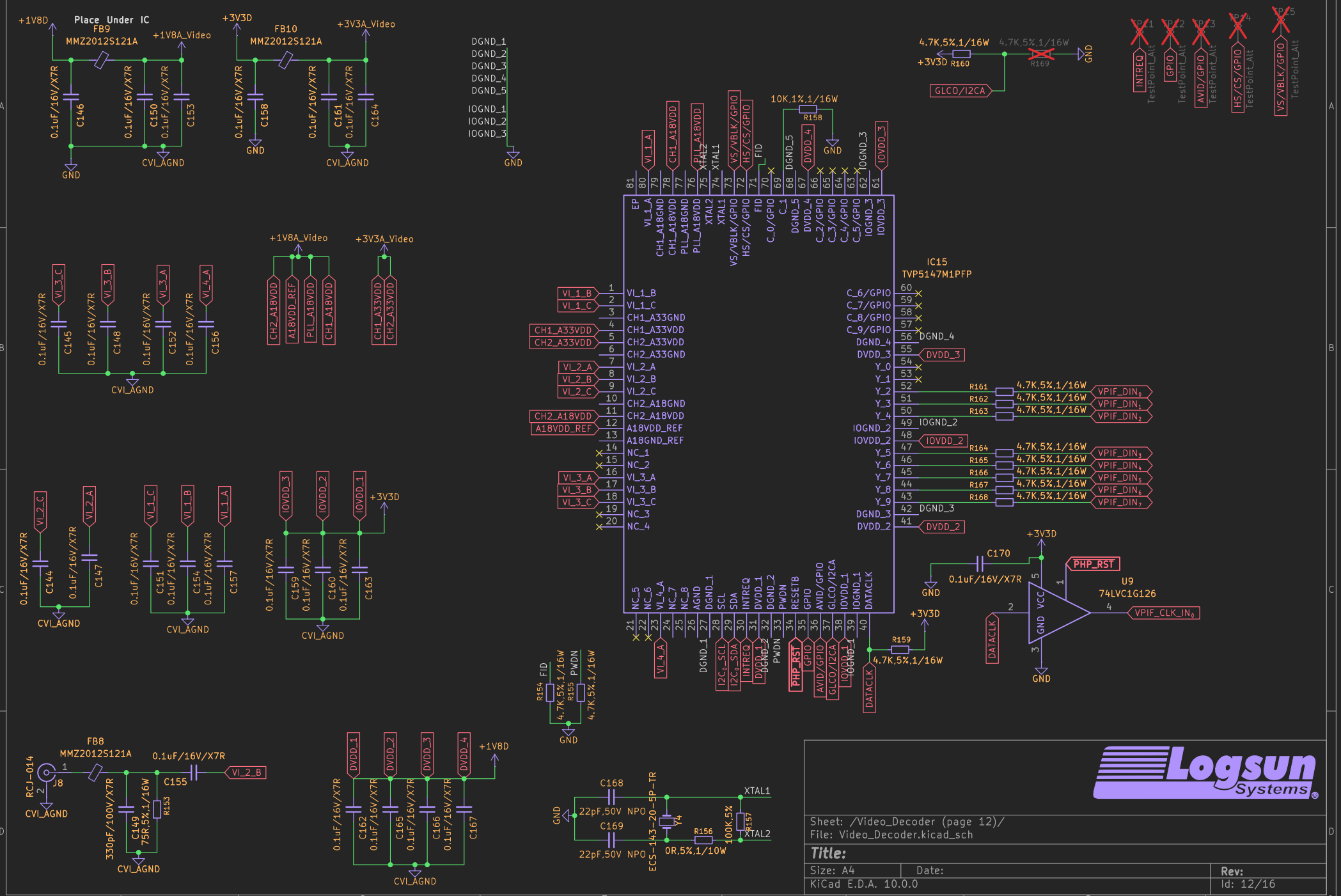
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 11/16



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File: Video_Decoder.kicad_sch

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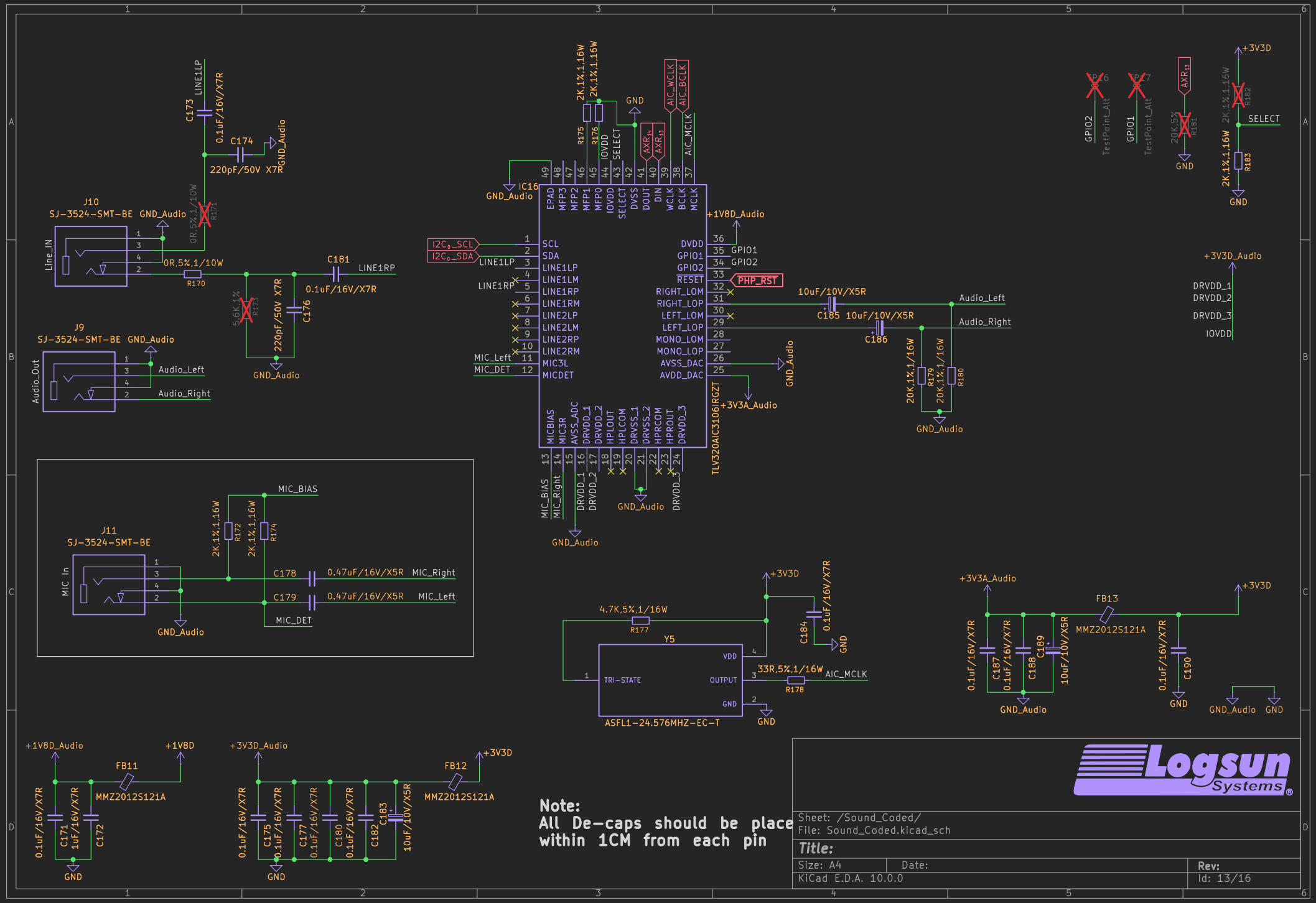
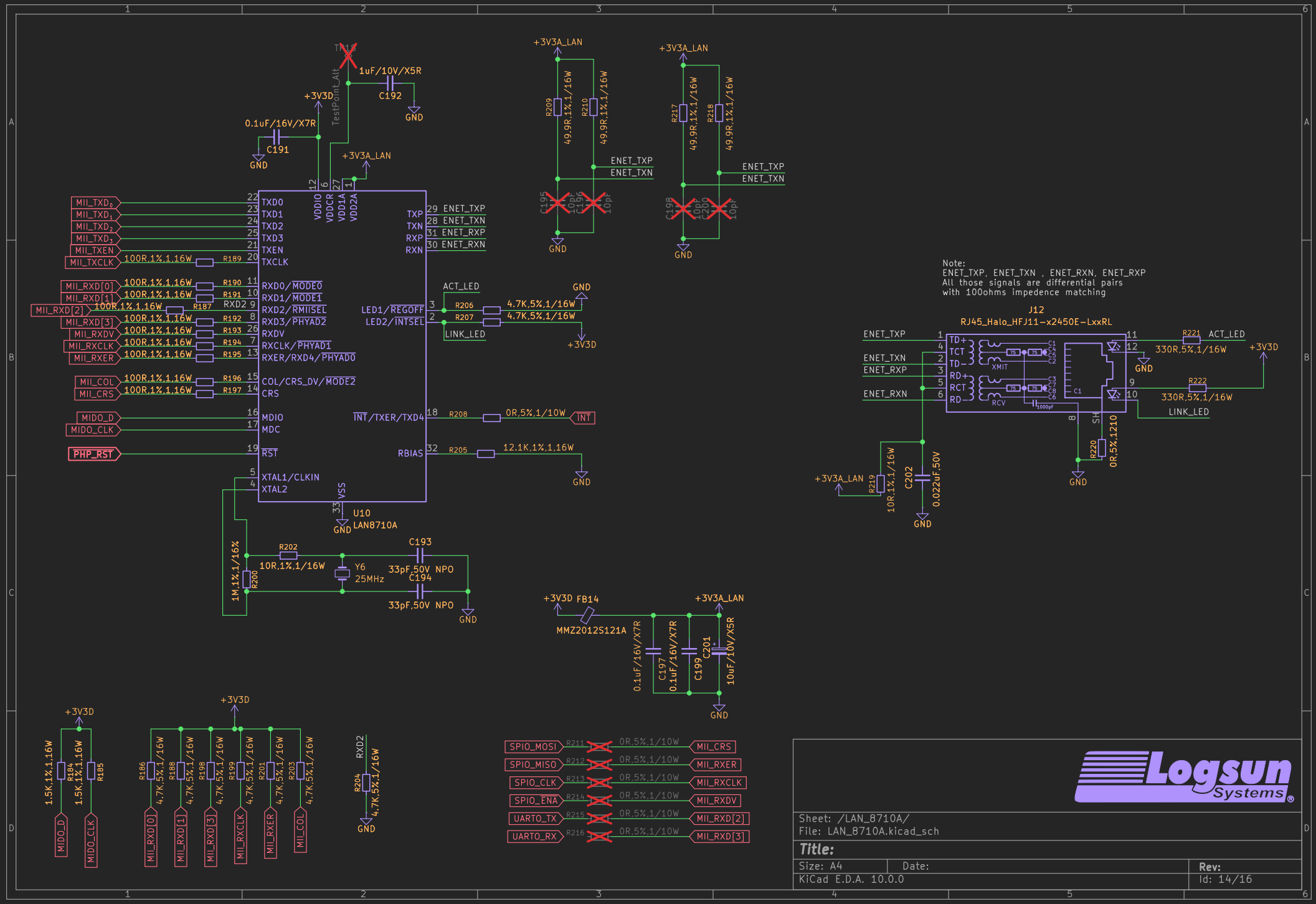
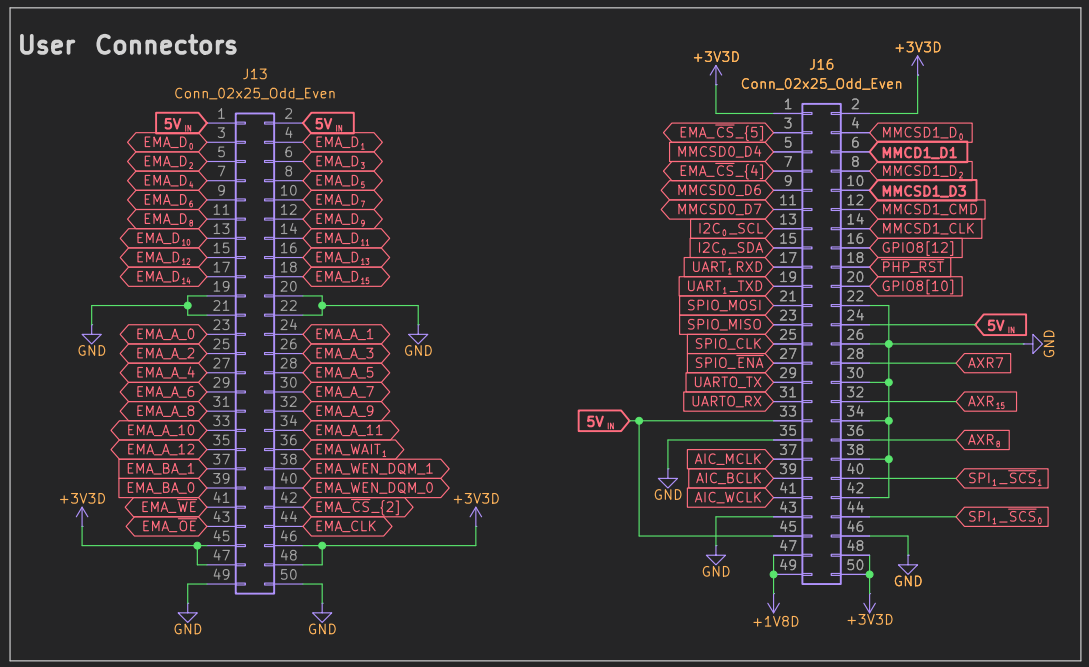
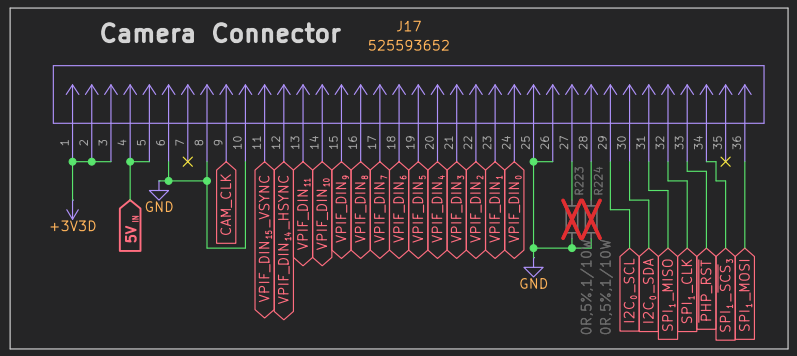
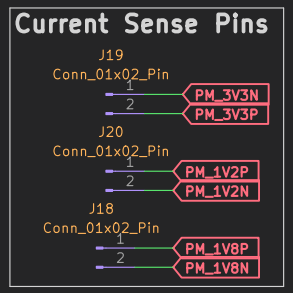
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 12/16

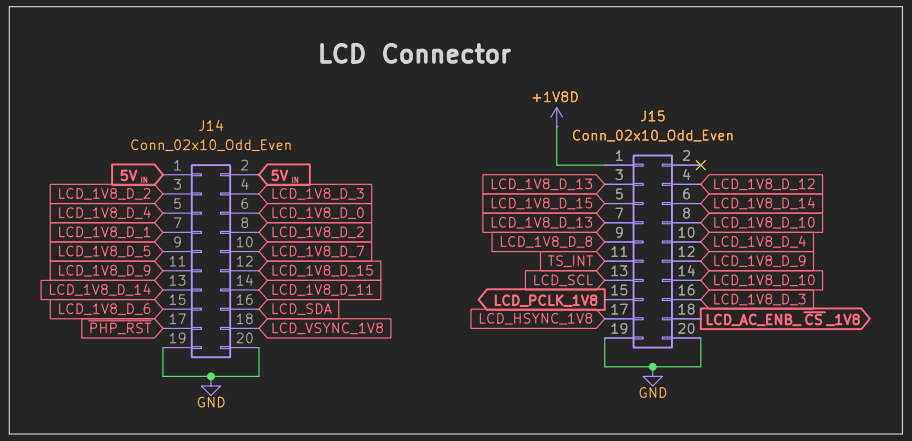

J10
SJ-3524-SMT-BE
GND_Audio
Line_IN
C173
LINE1LP
0.1uF/16V/X7R
C174
220pF/50V X7R
GND_Audio
0R,5%,1/10W
R171
0R,5%,1/10W
R170
C181
LINE1RP
0.1uF/16V/X7R
5.6K,1%
R175
220pF/50V X7R
C176
GND_Audio
J9
SJ-3524-SMT-BE
GND_Audio
Audio_Out
Audio_Left
Audio_Right
IC16
GND_Audio
I2C0_SCL
I2C0_SDA
LINE1LP
LINE1RP
MIC_Left
MIC_DET
SCL
SDA
LINE1LP
LINE1LM
LINE1RP
LINE1RM
LINE2LP
LINE2LM
LINE2RP
LINE2RM
MIC3L
MICDET
EPAD
MFP3
MFP2
MFP1
MFP0
IOVDD
SELECT
DVSS
DOUT
DIN
WCLK
BCLK
MCLK
2K,1%,1.16W
R175
2K,1%,1.16W
R176
GND
AIC_WCLK
AIC_BCLK
AXR14
AXR13
AIC_MCLK
+1V8D_Audio
DVDD
GPIO1
GPIO2
RESET
PHP_RST
RIGHT_LOM
RIGHT_LOP
LEFT_LOM
LEFT_LOP
MONO_LOM
MONO_LOP
AVSS_DAC
AVDD_DAC
GND_Audio
+3V3A_Audio
TLV320AIC3106IRGZT
MICBIAS
MIC3R
AVSS_ADC
DRVDD_1
DRVDD_2
HPLOUT
HPLCOM
DRVSS_1
DRVSS_2
HPRCOM
HPROUT
DRVDD_3
MIC_BIAS
MIC_Right
DRVDD_1
DRVDD_2
GND_Audio
GND_Audio
DRVDD_3
10uF/10V/X5R
C185
10uF/10V/X5R
C186
Audio_Left
Audio_Right
20K,1%,1/16W
R179
20K,1%,1/16W
R180
GND_Audio
GPIO2
TestPoint_Alt
GPIO1
TestPoint_Alt
AXR13
20K,5%
R181
GND
+3V3D
2K,1%,1.16W
R182
SELECT
2K,1%,1.16W
R183
GND
+3V3D_Audio
DRVDD_1
DRVDD_2
DRVDD_3
IOVDD
J11
SJ-3524-SMT-BE
MIC In
GND_Audio
2K,1%,1.16W
R172
2K,1%,1.16W
R174
MIC_BIAS
C178
0.47uF/16V/X5R
MIC_Right
C179
0.47uF/16V/X5R
MIC_Left
MIC_DET
+3V3D
4.7K,5%,1/16W
R177
C184
0.1uF/16V/X7R
GND
Y5
VDD
TRI-STATE
OUTPUT
GND
33R,5%,1/16W
R178
AIC_MCLK
GND
ASFL1-24.576MHZ-EC-T
+3V3A_Audio
0.1uF/16V/X7R
C187
0.1uF/16V/X7R
C188
C189
10uF/10V/X5R
FB13
MMZ2012S121A
0.1uF/16V/X7R
C190
+3V3D
GND_Audio
GND_Audio
GND
+1V8D_Audio
FB11
MMZ2012S121A
+1V8D
0.1uF/16V/X7R
C171
1uF/16V/X7R
C172
GND
+3V3D_Audio
FB12
MMZ2012S121A
+3V3D
0.1uF/16V/X7R
C175
0.1uF/16V/X7R
C177
0.1uF/16V/X7R
C180
0.1uF/16V/X7R
C182
C183
10uF/10V/X5R
GND
Note:
All De-caps should be place
within 1CM from each pin
Logsun Systems
Sheet: /Sound_Coded/
File: Sound_Coded.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 13/16


U10
LAN8710A
J12
RJ45_Halo_HFJ11-x2450E-LxxRL
Note:
ENET_TXP, ENET_TXN , ENET_RXN, ENET_RXP
All those signals are differential pairs
with 100ohms impedence matching
Logsun Systems
Sheet: /LAN_8710A/
File: LAN_8710A.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 14/16









GPIO0[8]

Logsun Systems

Sheet: /Connectors/
File: Connectors.kicad_sch

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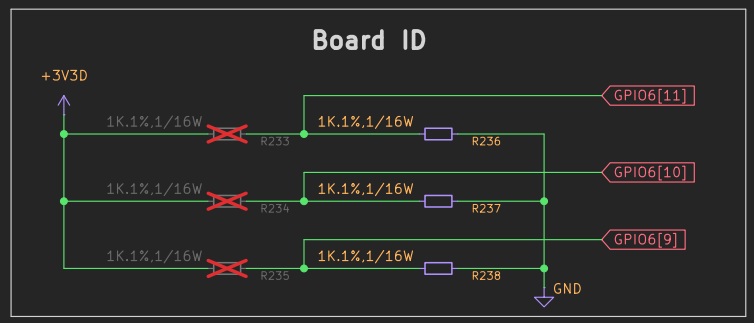
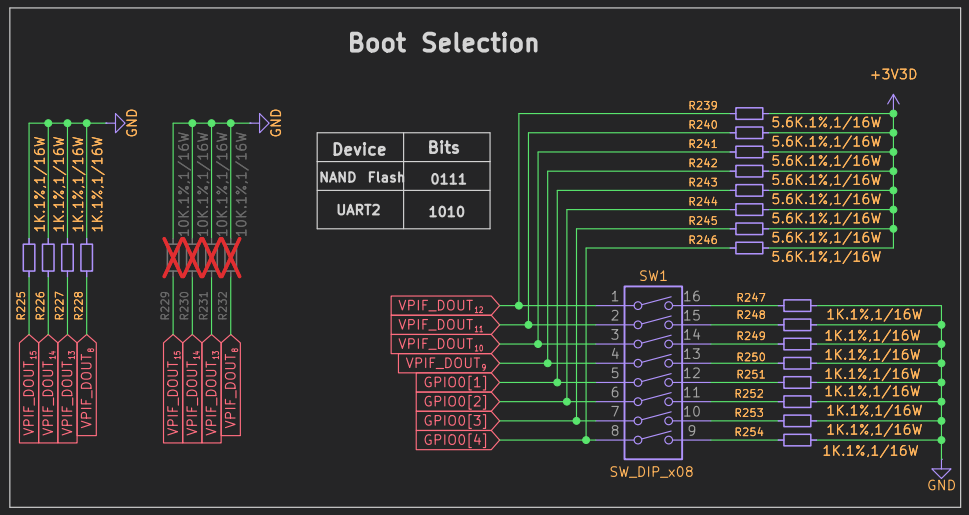
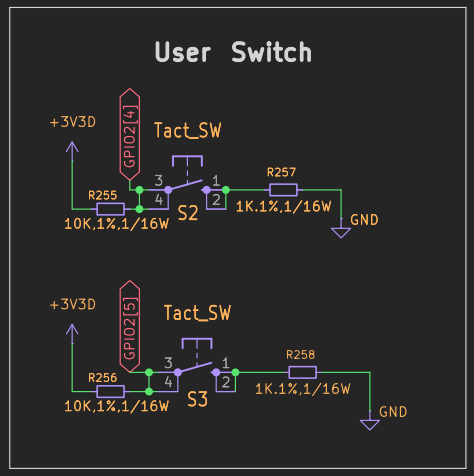
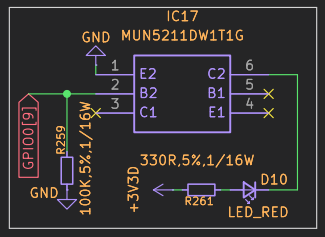
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 15/16

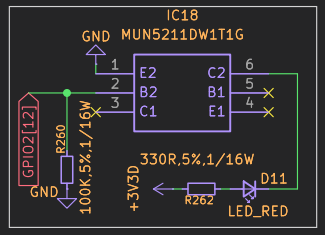




Device	Bits
NAND Flash	0111
UART2	1010







Logsun Systems

Sheet: /Mislleneous/
File: Mislleneous.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 16/16